Sai Kiran Pennam

+1 (512) 409 0668 | saikiran.pennam@icloud.com | www.linkedin.com/in/saikiranpennam
F1 STEM-OPT at Austin, TX

Education & Certificates

- Master of Science in Computer Science | Colorado State University Graduated 12 / 2024

Relevant coursework: Deep Learning, Big Data, Software Engineering, Computer & Quantitative Security

- AWS Certified Machine Learning Engineer 03 / 2025

Work History

Data Engineer, Rivian Systems, LLC, Remote *02 / 2026 – Present*

- Used Amazon S3 and AWS Glue to collect and organize machine-sensor files in one secure cloud location; replaced scattered spreadsheets with a reliable central data source.
- Built automated ETL pipelines with AWS Glue and Python; cleaned, transformed, and loaded daily sensor and maintenance data so analysts use it without manual preparation.
- Created data quality checks with AWS Glue and Amazon CloudWatch; detected missing duplicate, or unusual sensor readings before they affected dashboards or machine-learning models.
- Built Amazon QuickSight dashboards using curated Redshift data; gave operations leaders simple reports on machine downtime, production output, and maintenance trends.

AI Safety Researcher, Algoverse, Remote *05 / 2026 – 09 / 2026*

- Used Nvidia A100 GPUs to fine-tune Gemma 3 4B IT vision-language models on targeted datasets to study how narrow fine-tuning can cause unsafe or misaligned behavior.
- Reproduced a latent intervention method for text representations in VLMs; tested whether changing internal text-related directions could reduce or induce misaligned model behavior.
- Used vLLM to run high-throughput model inference during safety evaluations; made it practical to test many intervention, prompt, and model-output conditions.
- Combined MMSafetyBench results with LLM-as-a-judge and human review; checked whether apparent misalignment effects were real, consistent, and not caused by flawed or out-of-distribution evaluation inputs.
- Contributed to mechanistic interpretability* analysis comparing text- and image-conditioned representations; supported evidence that text-direction interventions, rather than vision-direction interventions, were causally responsible for the observed misaligned behavior.

* Anonymous manuscript on mechanistic interpretability in vision-language models. Accepted to InterpScience Workshop at NeurIPS. Camera-ready version in progress.

Data Scientist, TT Technologies, Remote *04 / 2025 – 01 / 2026*

- Used SQL and data analysis to combine website, ad, and purchase data; identified the marketing channels what brought the most paying customers.
- Applied customer segmentation to group shoppers by behavior, such as first-time buyers, repeat customer, and inactive customers; helped launch targeted emails that increased repeat purchases by 12%.
- Created a forecasting model using historical campaign and seasonal data; helped the team plan advertising budget and inventory for high-demand prices.

Graduate Research Assistant, CSU Ag Bio, Colorado *01 / 2024 – 12 / 2024*

- Helped researchers use plant pictures and light data to find disease early; used Python and Machine Learning to turn complex plant data into clear results.
- Used data preprocessing to clean and organize hyperspectral data from 122 sugar beet plants, making the dataset reliable for disease-detection experiments.
- Applied feature engineering and statistical analysis to select useful wavelength bands, helping models distinguish healthy plants from plants with soilborne disease.
- Built and tested machine learning models, including SVM, Random Forest, MLP, and ResNet50 to compare disease-identification and severity-estimation performance.
- Used model evaluation and adversarial edge-case testing to identify class-imbalance and out-of-distribution weaknesses, improving the reliability of the final research results.
- Created reproducible data pipelines with organized code, experiment tracking, and versioned workflows, helping collaborators repeat analyses and build on the project.

Projects

LLM Security Evaluation System – RAG pipelines & Behavioral Evaluations 02 / 2024 – 05 / 2024

- Built a RAG pipeline that helps an AI system look up trusted security information before answering; chose retrieval over fine-tuning because security knowledge detailed, traceable source information.
- Developed behavioral evaluations to check whether answers use the right retrieved information and stay true to it; measured retrieval, answer, faithfulness, and response quality.

Skills

Languages:	Python, C++, Bash/Shell scripting
ML & Eval Frameworks:	Scikit-learn, pytorch, JAX, huggingface-cli, transformers, opencv
LLM & RAG Tooling:	LLM APIs (Claude/OpenAI), vLLM, Vector databases, langchain, wandb
Data Analysis:	Pandas, numpy, matplotlib, seaborn
Cloud & Infrastructure:	AWS, Docker, FastAPI, Git
Databases:	PostgreSQL, MySQL